

MULTIPLE TESTING PROCEDURES FOR THE k-SAMPLE RUNS TEST

S.G. Reinach

University of Port Elizabeth, South Africa

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SUMMARY

In this paper simultaneous confidence intervals as well as multiple test procedures are derived for the number of runs in the k-sample runs problem. Some of these procedures depend on the upper α -point of the maximum of k equi-correlated unit normal random variables. Tables of critical values are constructed and presented. An application of these methods is illustrated by means of an example.

1. INTRODUCTION

Consider an N-vector $X' = (X_1, X_2, \dots, X_N)$ where $X_j \in \{A_i; i = 1, \dots, k\}$ for $j = 1, \dots, N$ and where $A_i, i = 1, \dots, k$ denote distinct abstract entities (the A_i 's are sometimes referred to as treatments). For $i = 1, \dots, k$ let n_i denote the number of times A_i appears in X (also referred to as replicates of i-th treatment). Thus $N = \sum_{i=1}^k n_i$. The elements of X are partitioned into sets of identical elements occurring consecutively. Each of these sets is termed a run. Let there be R such runs in X and let A_i admit R_i distinct runs.

As an illustrative example, consider

$$X = (\underline{A_3}, \underline{A_3}, \underline{A_2}, \underline{A_1}, \underline{A_1}, \underline{A_1}, \underline{A_2}, \underline{A_2}, \underline{A_1})$$

In this case $N = 9$, $n_1 = 4$, $n_2 = 3$, $n_3 = 2$ and the $R = 5$ runs are underlined with $R_1 = 2$, $R_2 = 2$, $R_3 = 1$.

The null hypothesis being considered is that the abstract entities occur randomly, i.e. each of the $\binom{N}{n_1 n_2 \dots n_k}$ possible arrangements of these abstract entities is regarded as equally likely.

Various alternatives can be considered but in this paper we are only considering the

following two one-sided alternatives, namely:

H_{a1} : Too few runs of one or more of the abstract entities A_i , $i = 1, \dots, k$.

In this case we suspect definite grouping of one or more of the entities. Thus we refer to H_{a1} as the clustering hypothesis.

H_{a2} : Too many runs of one or more of the abstract entities A_i , $i = 1, \dots, k$.

In this case we suspect some sort of repeated alternating pattern of one or more of the A_i , $i = 1, \dots, k$. Thus we refer to H_{a2} as the hypothesis of evenly spread entities.

The theory of the case $k = 2$ (two-sample test) was originally derived by Whitworth (1886) and revived by Stevens (1939), Wald & Wolfowitz (1940) and Mood (1940). Swed and Eisenhart (1943) calculated the probability distribution of the total number of runs R in the case $n_1 \leq n_2 \leq 20$. These tables or parts thereof have been reproduced in several textbooks. If $k = 2$ and N large ($N \geq 20$) the normal approximation for the distribution of R can be used to test H_0 .

When $k = 2$ it is immaterial whether R_1 or R is used as test statistic. The reason for this is that if H_0 is rejected, both abstract entities are clustered or both have too many runs.

The theory for the case $k \geq 3$ was first developed by Mood (1940) who derived test statistics based on the total number of runs, R , as well as one based on the number of runs R_i , $i = 1, \dots, k$.

Barton & David (1957) also developed the theory for the total number of runs. Hiraga et al (1954) calculated exact 5% critical values of R , for $n_1 \leq n_2 \leq n_3 \leq 10$. Critical values were also calculated by using the normal approximation for particular values in the range $10 \leq n_1 \leq n_2 \leq n_3 \leq 30$.

The outcome of the test based on the total number of runs R , is sensitive to deviations from H_0 only if there is a tendency for most of the abstract entities $\{A_i, i = 1, \dots, k\}$ either to cluster or to be evenly spread. If some cluster and others are evenly spread, the test based on R is not very sensitive for rejecting H_0 as illustrated by the following example. Consider the sequence of A_i 's:

$$\frac{A_1 A_1 A_1 A_1 A_1 A_1 A_1 A_1 A_1 A_1 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3 A_2 A_3}{A_2 A_3}$$

Thus $k = 3$; $n_1 = n_2 = n_3 = 10$; $R_1 = 1$; $R_2 = R_3 = 10$ and $R = 21$.

From the tables of Hiraga (1954) the 5% critical value for R is $R(0.05) = 16$. Thus R is not significant and we do not reject H_0 .

However, if a Bonferroni procedure is followed, namely making all pairwise comparisons of the abstract entities A_i with the appropriate test when $k = 2$, at an individual test level such that the overall confidence remains at least $1 - \alpha$, it is found that A_1 and A_2 as well as A_1 and A_3 are clustered when considered together.

The test statistic of k samples, developed by Mood (1940), and which is based on each R_i , is better for detecting clustering if only some of the A_i 's, $i = 1, \dots, k$, are clustered. However, like the F-test in the analysis of variance, the test statistic developed by Mood provides an overall test but does not distinguish the abstract entities responsible for the rejection of H_0 .

In this paper two multiple comparison procedures, based on the individual R_i or linear combinations of them, are derived. Critical values needed for these procedures are presented in Tables 3 and 4. The exact probabilities were also calculated for Mood's (1940) test for the case $k = 3$; $n_1 = n_2 = n_3 = 2, 3, 4$ and 5. These values are presented in Table 1.

2. Preliminaries

In terms of the quantities defined in Section 1, Mood (1940) proved the following under H_0 :

$$(1) E(R_i) = \frac{n_i(N - n_i + 1)}{N}$$

$$(2) \text{var}(R_i) = \frac{n_i(n_i - 1)(N - n_i + 1)(N - n_i)}{N^2(N - 1)}$$

Consider now:

$$(3) U_i = \frac{R_i - E(R_i)}{\sqrt{\text{var}(R_i)}}$$

From Mood (1940) it follows that the U_i 's, $i = 1, \dots, k$ are asymptotically (for $n_i \rightarrow \infty$) normally distributed with zero means, unit variances and covariances given by

$$(4) \sigma_{ij} = \frac{n_i(n_i - 1)n_j(n_j - 1)}{[n_i(n_i - 1)(N - n_i + 1)(N - n_i)n_j(n_j - 1)(N - n_j + 1)(N - n_j)]^{1/2}}$$

Mood further proved that, under H_0 ,

$$(5) Q = \underline{U}' \underline{\Sigma}^{-1} \underline{U},$$

where the components of $\underline{U}$ are defined in (3) and where $\mathfrak{Z}$ is the matrix with diagonal elements 1 and off-diagonal elements as in (4), is asymptotically distributed according to the χ^2 distribution with k degrees of freedom.

3. Test Statistics

3.1 Derivation of simultaneous confidence intervals based on the U_i

These intervals are based on the method of multiple comparison used by Scheffé (1951) for the F-test.

Lemma:

Given an arbitrary vector x and a positive definite matrix $B(p \times p)$, then, for any constant k ,

$$\underline{x}' B \underline{x} \leq k^2 \Leftrightarrow |\underline{c}' \underline{x}| \leq k \sqrt{\underline{c}' B^{-1} \underline{c}} \text{ for all vectors } \underline{c}.$$

This lemma enunciates the principle that all points inside an ellipsoid lie between all of its parallel planes of support. See for example Ferguson (1967), p. 282.

Theorem:

Asymptotically, if $n_i \rightarrow \infty$ in such a way that $\lim(n_i/N) > 0$, $i = 1, \dots, k$, the probability is $(1-\alpha)$ that

$$(6) \quad |\underline{c}' \underline{U}| \leq \sqrt{\chi_k^2(\alpha)} \sqrt{\underline{c}' \Sigma \underline{c}} \text{ for all } \underline{c},$$

where $\underline{U}' = (U_1, \dots, U_k)$ is defined in (3), $\mathfrak{Z}$ is the matrix with diagonal elements 1 and off-diagonal elements defined in (4) and $\chi_k^2(\alpha)$ is the upper α -point of the χ^2 distribution with k degrees of freedom.

Proof:

From (5) we have that asymptotically (for $n_i \rightarrow \infty$ whilst $n_i/N > 0$) the probability is $(1-\alpha)$ that

$$\underline{U}' \mathfrak{Z}^{-1} \underline{U} \leq \chi_k^2(\alpha).$$

Since $\mathfrak{Z}$ is positive definite (see Mood (1940)), $\mathfrak{Z}^{-1}$ exists and (6) follows directly from the lemma.

Special case:

A critical α -point R_i^α can now be obtained for each R_i in the special case $n_i = n$,

$i = 1, \dots, k$. Thus by substituting from equations (1), (2) and (3) in (6) we get:

$$(7) R_i^\alpha = \frac{(nk-n+1)}{k} - \frac{1}{2} - \sqrt{\chi_k^2(\alpha)} \sqrt{\frac{(n-1)(k-1)(nk-n+1)}{k^2(nk-1)}}$$

Equation (7) gives the lower critical bound for the clustering alternative and the $\frac{1}{2}$ is a standard continuity correction. An observed value of $R_i \leq R_i^\alpha$ is declared significant at the α level.

3.2 A multiple testing procedure based on individual R_i 's

As can be seen from (7) a critical value can be obtained for individual R_i , $i = 1, \dots, k$, by means of the procedure derived in Section 3.1. In this section a more efficient test is suggested for the special case where individual abstract entities are tested for deviations from H_0 .

It is possible to derive the distribution of the tests for any n_i . However, since it is not feasible to compute tables for all possible combinations, the derivation is given for the special case $n_i = n$ only.

Since the U_i 's are asymptotically jointly normally distributed (see Mood (1940)) with unit variances and common correlation

$$(8) \rho = \frac{(n-1)}{(k-1)(nk-n+1)}, \text{ (from (4))},$$

we can now approximate the α -point, U^α , for the one-sided alternative, by

$$(9) U^\alpha = m_{k(\rho)}^\alpha.$$

For the clustering alternative $m_{k(\rho)}^\alpha$ is the (lower) α -point of the minimum of k equally correlated unit normal variables and thus negative for the usual values of α .

Substitution from (1), (2) and (3) in (9) yields:

$$(10) R_{k(\rho)}^\alpha = \frac{(nk-n+1)}{k} - \frac{1}{2} + m_{k(\rho)}^\alpha \sqrt{\frac{(n-1)(k-1)(nk-n+1)}{k^2(nk-1)}}$$

where the $\frac{1}{2}$ is a continuity correction.

Any observed $R_i \leq R_{k(\rho)}^\alpha$ will result in H_0 being rejected for the specific entity at the α level of significance and we conclude that the entity is clustered.

If the one-sided alternative hypothesis, namely evenly spread runs, is considered, equation (10) becomes:

$$R_{k(\rho)}^{\alpha*} = \frac{(nk-n+1)}{k} + \frac{1}{2} + m_{k(\rho)}^{\alpha*} \sqrt{\frac{(n-1)(k-1)(nk-n+1)}{k^2(nk-1)}}$$

In this case $m_{k(\rho)}^{\alpha*} = -m_{k(\rho)}^{\alpha}$ is the $(1-\alpha)$ -point (or upper α -point) of the maximum of k equally correlated unit normal variables.

4. Applicability of the Normal Approximation

The exact probabilities of the runs test defined by Mood (1940) were obtained for $k = 3$ and $n = 2, 3, 4$ and 5 , where n is the number of times the abstract entities occurred (alternatively called the number of observations per treatment). The results obtained are given in Table 1 together with the probabilities obtained from the χ^2 distribution with 3 degrees of freedom. It is observed that the exact probabilities (for $n_1 = n_2 = n_3 = 5$) are (except in the case of one critical value) less than the corresponding χ^2 probabilities. The use of the χ^2 approximation thus seems to be conservative and consequently safe to use.

Some investigation was also done on the marginal distribution of R_1 , the number of runs of the first treatment, to see whether it supports the above recommendation. It should be noted that the marginal distribution is the same whether $k = 2$ or more. In this case, $k > 2$, the sum of all the observations in the remaining treatments ($n_2 + \dots + n_k$) can be considered as the number of replicates forming a second treatment.

With the aid of the formulae for the marginal distribution of R_1 given in Mood (1940) the exact cumulative probabilities as well as the probabilities corresponding to the normal approximation (with a continuity correction of $\frac{1}{2}$) were calculated for different values of n_1, n_2 and r_1 , for the cases where the probabilities are in the vicinity of 0.05. These results are summarised in Table 2.

From Table 2 it can be seen that for samples as small as $n_1 = n_2 = 7$ and $n_1 = 6; n_2 = 12$ the normal approximation probabilities are conservative when compared to the exact probabilities. It thus seems reasonable to use the normal approximation to obtain probabilities for the runs test for samples with $n_1 \geq 6$.

5. Construction of Tables for $m_{k(\rho)}^{\alpha}$

Gupta (1963) constructed probability tables for the multivariate normal distribution with equal correlations. These tables are for $k = 1$ to 12, $H = -3.50$ to 3.50 and various values of ρ (where H is a percentile point of the maximum of k equally correlated normal random variates).

To apply equation (9), the critical values of $m_{k(\rho)}^{\alpha}$ are needed for numerous ρ values not tabled by Gupta.

Kendall (1942) gave a method to integrate the multivariate normal distribution with the aid of an infinite series. This method was used to construct the critical values

$m_{k(\rho)}^\alpha$ in Table 3 for various values of n and k .

Theoretically it is possible to obtain probabilities for unequal n_i . However, this is not practically feasible.

Only equal numbers $n_i = n$ are considered and thus all ρ 's were equal for a given n and k .

When $k = 2$ the value of ρ (in (8)) varies from 0,33 for $n = 2$ up to 0,961 for $n = 50$. The result is that $m_{k(\rho)}^\alpha$ also varies for $n = 2$ to $n = 50$. The correlation coefficients varies from 0,100 for $n = 2$ to 0,241 for $n = 50$ when $k = 3$, and in general they may be checked by using (8). The variation becomes smaller as k increases. From $k = 5$ there is only a change of two units in the third decimal place of $m_{k(\rho)}^\alpha$ when n varies from 2 to 50. As a result the values of $m_{k(\rho)}^\alpha$ are given only for $n = 10$.

In Table 3 the lower critical values of $m_{k(\rho)}^\alpha$ are given for $\alpha = 0,005; 0,01; 0,025; 0,05$ and $0,10$; $k = 2, 3, \dots, 10$ and $n = 2(1)10(5)20(10)50$. These values can be used for direct application of equations (10) and (11).

With the aid of the values in Table 3 the critical $R_{k(\rho)}^\alpha$ (equation (10)) were calculated for the clustering alternative and the results are summarised in Table 4. Any calculated R_i value $\leq R_{k(\rho)}^\alpha$ will lead to rejection of H_0 . The values in parentheses in Table 4 are the critical $R_{k(\rho)}^{\alpha*}$ (equation (11)) values for the evenly spread alternative. Any R_i value $\geq R_{k(\rho)}^{\alpha*}$ is declared significant.

The values in Table 4 were rounded off in such a way that the true significance level is always $\leq \alpha$.

6. Comparing the Efficiency of Two Procedures

In this section we show that our multiple testing procedure detects differences more readily than the simultaneous confidence procedure for the case where individual R_i 's are considered and all $n_i = n$.

Comparing equations (7) and (10) it is seen that the difference in the critical R

values lies only in the values of $m_{k(\rho)}^\alpha$ and $\sqrt{\chi_k^2(\alpha)}$.

The smaller of $|m_{k(\rho)}^{\alpha/2}|$ and $\sqrt{\chi_k^2(\alpha)}$ will yield the larger R_i^α value and will thus more readily reject H_0 .

Remarks

- (a) It should be noted for the two tests to be at a comparable level of significance, the test level for $m_{k(\rho)}^\alpha$ is $\alpha/2$, it being a one-sided test and for χ_k^2 it is α , the latter

being a two-sided test.

- (b) Testing for the alternative H_{a1} of clustering, the negative side of the multivariate normal distribution is used and thus the absolute value of $m_{k(\rho)}^{\alpha/2}$ is compared with $+\sqrt{\chi_k^2(\alpha)}$.

The values of $m_{k(\rho)}^{\alpha/2}$ are obtained from Table 3. For a given k at level $\alpha/2$ the largest value of $m_{k(\rho)}^{\alpha/2}$ is used for comparison with $\sqrt{\chi_k^2(\alpha)}$. The results are summarised in Table A.

TABEL A

A comparison of the values of $|m_{k(\rho)}^{\alpha/2}|$ and $\sqrt{\chi_k^2(\alpha)}$ at $\alpha = 0,05$ for different values of k

k	3	4	5	6	7	8	9	10
$ m_{k(\rho)}^{0,025} $	2,388	2,493	2,571	2,634	2,686	2,730	2,769	2,803
$\sqrt{\chi_k^2(0,05)}$	2.795	3,080	3,327	3,549	3,751	3,938	4,113	4,279

From Table A it is seen that the values used in the multiple testing procedure are smaller than the simultaneous confidence interval values. It will thus result in large critical R_i^α value which means that H_0 will be rejected sooner when testing individual treatments.

7. Application

The procedures described are now illustrated by means of an hypothetical example

Example:

A group of 30 foreign students from three different countries, say, Australia (10), Canada (10) and England (10), studying at a university in the U.S.A., were called upon to have a medical check-up as well as other tests performed on them. They were left standing in a room to become acquainted and then asked to take seats in the next room where chairs were set in a row next to each other. Our problem is whether the resulting seating arrangement is random or whether some form of clustering occurred. The seating arrangement that occurred was as follows, where the symbols used have an obvious meaning

aaa e ccc ee a a a ccc ee aaaa c e c e c e c e

Here we have $n_a = n_e = n_c = 10$; $N = 30$; $R_a = 4$; $R_e = 8$; $R_c = 6$.

From (1) : $E(R_i) = 7$.

From (3) : $U_a = -2,4928$; $U_e = 0,8309$; $U_c = -0,8309$

Although it is not required for our analysis, we calculate the overall test statistic defined by Mood (equation (5)). Thus

$$\mathbf{Z} = \begin{vmatrix} 1,0000 & 0,2143 & 0,2143 \\ 0,2143 & 1,0000 & 0,2143 \\ 0,2143 & 0,2143 & 1,0000 \end{vmatrix} \quad \text{and} \quad \mathbf{Z}^{-1} = \begin{vmatrix} 1,0818 & -0,1909 & -0,1909 \\ -0,1909 & 1,0818 & -0,1909 \\ -0,1909 & -0,1909 & 1,0818 \end{vmatrix}$$

Substitution in (5) yields $Q = 8,798$ and since $\chi_3^2(0,05) = 7,815$ the result is significant at the 5% level.

- (i) We now illustrate the use of equation (6). Consider any pairwise comparison, for example $\mathbf{c}' = (1 \ -1 \ 0)$. Thus $\mathbf{c}' \mathbf{Z} \mathbf{c} = 1,5714$. Due to the fact that $n_i = n$ is for all i , all pairwise comparisons will yield the same value for $\mathbf{c}' \mathbf{Z} \mathbf{c}$.

Let $\hat{\psi} = |\mathbf{c}' \mathbf{U}| = |U_a - U| = 3,3237$.

Indicating the minimum value of $\hat{\psi}$ to be significant at the α level by $\text{LSD}(\hat{\psi})$ we get

$$\begin{aligned} \text{LSD}(\hat{\psi}) &= \text{LSD}|U_a - U_e| = \sqrt{\chi_k^2(\alpha) \mathbf{c}' \Sigma \mathbf{c}}, \quad \alpha = 0,05; k = 3. \\ &= \sqrt{7,815 \times 1,5714} \\ &= 3,504. \end{aligned}$$

Since $\hat{\psi} < \text{LSD}(\hat{\psi})$ the contrast considered is not declared significant at an overall 5 percent level of significance. Equation (6) can also be used to test any individual U_i and thus R_i .

However, we will make use of equation (7).

- (ii) To test R_i with the aid of equation (7):

$$R_i^{0,05} = 3,14$$

We see that $R_a = 4$ and is thus not significant at the 5 percent level.

- (iii) To test any R_i with the aid of equation (10), we find:

$$R_{3(\rho)}^{0,05} = 3,96 \text{ (from equation (10) and Table 3).}$$

From Table 4 we obtain a value of 3 for $R_{3(\rho)}^{0,05}$. Thus once again, no R_i is significant at the 5 percent level.

8. Concluding Remarks

- (i) It should be noted that if a U_i value is negative, it indicates the tendency of the i -th abstract entity to cluster whilst if it is positive, it shows a tendency towards a more evenly spread distribution of the A_i 's.
- (ii) Equation (6) is applicable for any n_i whilst (10) is only applicable if all $n_i = n$ since no tables exist for different n_i .
- (iii) To test the correctness of the calculations of Table 3, the values obtained on $\rho = 0,1$ and $k = 2,3, \dots, 10$ were compared with those tabulated by Gupta (1963). The correspondence was to 5 digits.

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TABLE 1

Critical values of the k-sample runs test for $k = 3$ and $n_1 = n_2 = n_3 = 2, 3, 4$ and 5 .

Sample Sizes			Critical Value	Exceedance Probabilities	
n_1	n_2	n_3		Exact	χ^2 approximation
2	2	2	5,000	0,067	0,172
			4,583	0,267	0,204
3	3	3	10,667	0,007	
			8,000	0,061	0,046
			6,667	0,132	0,082
			4,667	0,218	0,197
4	4	4	10,725	0,013	0,017
			8,800	0,022	0,032
			8,525	0,038	0,036
			7,700	0,052	0,052
			6,325	0,107	0,097
			4,400	0,230	0,221
5	5	5	10,267	0,015	
			8,750	0,017	0,032
			8,517	0,026	0,036
			8,167	0,043	0,042
			7,817	0,049	0,050
			6,767	0,063	0,078
			6,067	0,088	0,107
			5,600	0,097	0,133
			5,017	0,132	0,172
			4,900	0,192	0,179

TABLE 2

A comparison of the exact and the normal approximation probabilities for the number of runs in the marginal distribution of the two sample runs test.

The probabilities are $P(R_1 \leq r_1)$.

Number of Runs	Prob. based on	$n_1 = 6$	$n_1 = 7$	$n_1 = 8$	$n_1 = 9$	$n_1 = 10$	$n_1 = 6$	$n_1 = 7$	$n_1 = 8$
		$n_2 = 6$	$n_2 = 7$	$n_2 = 8$	$n_2 = 9$	$n_2 = 10$	$n_2 = 12$	$n_2 = 14$	$n_2 = 16$
1	Normal	0,0125	0,0046	0,0016	0,0006	0,0002	0,0011	0,0002	0,0001
	Exact	0,0076	0,0023	0,0007	0,0002	0,0001	0,0007	0,0001	0,0000
2	Normal	0,1314	0,0594	0,0256	0,0108	0,0043	0,0233	0,0062	0,0016
	Exact	0,1212	0,0513	0,0203	0,0076	0,0028	0,0217	0,0055	0,0013
3	Normal	—	0,3015	0,1650	0,0833	0,0401	0,1841	0,0668	0,0217
	Exact	—	0,2961	0,1570	0,0767	0,0350	0,1758	0,0642	0,0207
4	Normal	—	—	—	—	0,1910	—	0,3085	0,1379
	Exact	—	—	—	—	0,1850	—	0,2990	0,1340

TABLE 3

Critical values $m_{k(\rho)}^\alpha$ of the multivariate normal distribution $1 - \alpha = \int_{m_{k(\rho)}^\alpha}^\infty \dots \int_{m_{k(\rho)}^\alpha}^\infty dF$

$\alpha = 0,005$									
n \ k	2	3	4	5	6	7	8	9	10
2	-2,802	-2,933	-3,022	-3,089	-3,143				
3	-2,794	-2,933	-3,022	-3,089	-3,143				
4	-2,786	-2,932	-3,022	-3,089	-3,143				
5	-2,778	-2,932	-3,022	-3,089	-3,143				
6	-2,771	-2,932	-3,022	-3,089	-3,143				
7	-2,764	-2,932	-3,022	-3,089	-3,143				
8	-2,758	-2,932	-3,022	-3,089	-3,143				
9	-2,753	-2,932	-3,022	-3,089	-3,143				
10	-2,748	-2,931	-3,022	-3,089	-3,143	-3,188	-3,226	-3,260	-3,290
15	-2,729	-2,931	-3,021	-3,089					
20	-2,715	-2,931	-3,021	-3,089					
30	-2,696	-2,931	-3,021	-3,089					
40	-2,684	-2,931	-3,021	-3,089					
50	-2,676	-2,931	-3,021	-3,089	-3,143				
$\alpha = 0,01$									
2	-2,568	-2,710	-2,805	-2,876	-2,933				
3	-2,558	-2,709	-2,805	-2,876	-2,933				
4	-2,547	-2,709	-2,805	-2,876	-2,933				
5	-2,538	-2,708	-2,804	-2,876	-2,933				
6	-2,530	-2,708	-2,804	-2,876	-2,933				
7	-2,522	-2,708	-2,804	-2,876	-2,933				
8	-2,516	-2,707	-2,804	-2,876	-2,933				
9	-2,510	-2,707	-2,804	-2,876	-2,933				
10	-2,504	-2,707	-2,804	-2,876	-2,933	-2,981	-3,021	-3,057	-3,089
15	-2,485	-2,707	-2,804	-2,876	-2,933				
20	-2,467	-2,706	-2,804	-2,876					
30	-2,449	-2,706	-2,804	-2,876					
40	-2,438	-2,706	-2,804	-2,876	-2,933				
50	-2,430	-2,706	-2,804	-2,876	-2,933				

TABLE 3 (Continued)

$\alpha = 0,025$									
n \ k	2	3	4	5	6	7	8	9	10
2	-2,227	-2,388	-2,493	-2,571	-2,634				
3	-2,212	-2,386	-2,492	-2,571	-2,634				
4	-2,199	-2,385	-2,492	-2,571	-2,633				
5	-2,187	-2,384	-2,491	-2,571	-2,633				
6	-2,177	-2,383	-2,491	-2,570	-2,633				
7	-2,167	-2,383	-2,491	-2,570	-2,633				
8	-2,159	-2,382	-2,491	-2,570	-2,633				
9	-2,152	-2,382	-2,491	-2,570	-2,633				
10	-2,146	-2,382	-2,491	-2,570	-2,633	-2,686	-2,730	-2,769	-2,803
15	-2,123	-2,381	-2,491	-2,570					
20	-2,106	-2,381	-2,490	-2,570					
30	-2,086	-2,380	-2,490	-2,570					
40	-2,074	-2,380	-2,490	-2,570					
50	-2,066	-2,380	-2,490	-2,570	-2,633				
n	$\alpha = 0,05$								
2	-1,935	-2,116	-2,231	-2,317	-2,385				
3	-1,916	-2,113	-2,230	-2,316	-2,385				
4	-1,900	-2,111	-2,229	-2,316	-2,384				
5	-1,886	-2,110	-2,229	-2,316	-2,384				
6	-2,874	-2,109	-2,228	-2,315	-2,384				
7	-1,863	-2,108	-2,228	-2,315	-2,384				
8	-1,854	-2,107	-2,228	-2,315	-2,384				
9	-1,846	-2,107	-2,228	-2,315	-2,384				
10	-1,839	-2,107	-2,228	-2,315	-2,384	-2,440	-2,448	-2,530	-2,567
15	-1,813	-2,105	-2,227	-2,315					
20	-1,795	-2,105	-2,227	-2,315					
30	-1,774	-2,104	-2,227	-2,315					
40	-1,762	-2,104	-2,227	-2,315					
50	-1,753	-2,104	-2,227	-2,315	-2,384				

TABLE 3 (Continued)

$\alpha = 0,10$										
n	k	2	3	4	5	6	7	8	9	10
2		-1,603	-1,809	-1,938	-2,033	-2,108				
3		-1,577	-1,804	-1,936	-2,032	-2,108				
4		-1,556	-1,801	-1,935	-2,031	-2,107				
5		-1,539	-1,799	-1,934	-2,031	-2,107				
6		-1,525	-1,797	-1,933	-2,031	-2,107				
7		-1,513	-1,796	-1,933	-2,030	-2,107				
8		-1,502	-1,796	-1,933	-2,030	-2,106				
9		-1,493	-1,795	-1,933	-2,030	-2,106				
10		-1,485	-1,794	-1,932	-2,030	-2,106				
15		-1,456	-1,793	-1,932	-2,030					
20		-1,437	-1,792	-1,931	-2,030					
30		-1,414	-1,791	-1,931	-2,029					
40		-1,401	-1,790	-1,931	-2,029					
50		-1,392	-1,790	-1,931	-2,029	-2,106				

TABLE 4

Critical $R_{k(\rho)}^{\alpha}$ values for the k-sample runs test for a clustering alternative. Observed values $R_i \leq R_{k(\rho)}^{\alpha}$ are declared significant. Values in parentheses are the one-sided critical values $R_{k(\rho)}^{\alpha*}$ for an alternative of even spread. Observed values $R_i \geq R_{k(\rho)}^{\alpha*}$ are declared significant.

$\alpha = 0,01$								
n \ k	3	4	5	6	7	8	9	10
2	-(-)	-(-)	-(-)	-(-)	-	-	-	-
3	-(-)	-(-)	-(-)	-(-)	-	-	-	-
4	-(-)	-(-)	1(-)	1(-)	1	1	1	1
5	-(-)	1(-)	1(-)	1(-)	1	2	2	2
6	1(-)	1(-)	2(-)	2(-)	2	2	2	2
7	1(-)	2(-)	2(-)	2(-)	3	3	3	3
8	2(-)	2(-)	3(-)	3(-)	3	4	4	4
9	2(-)	3(-)	3(-)	4(-)	4	4	5	5
10	3(-)	4(-)	4(-)	4(-)	5	5	5	5
11	3(-)	4(-)	5(-)	5(-)	5	6	6	6
12	4(-)	5(-)	5(-)	6(-)	6	6	7	7
13	4(-)	5(-)	6(-)	6(-)	7	7	8	8
14	5(-)	6(-)	7(-)	7(-)	8	8	8	9
15	5(15)	6(-)	7(-)	8(-)	8	9	9	9
16	6(16)	7(-)	8(-)	9(-)	9	9	10	10
17	6(17)	8(-)	9(-)	9(-)	10	10	11	11
18	7(18)	8(-)	9(-)	10(-)	11	11	11	12
19	7(19)	9(-)	10(-)	11(-)	11	12	12	12
20	8(19)	10(-)	11(-)	11(-)	12	13	13	13
22	9(21)	11(-)	12(-)	13(-)	14	14	14	15
24	10(22)	12(24)	13(-)	14(-)	15	16	16	16
26	11(24)	13(26)	15(-)	16(-)	17	17	18	18
28	13(25)	15(28)	16(-)	17(-)	18	19	19	20
30	14(27)	16(29)	18(-)	19(-)	20	20	21	21
32	15(29)	17(31)	19(-)	20(-)	21	22	22	22
34	16(30)	19(33)	20(34)	22(-)	23	23	24	25
36	17(32)	20(35)	22(36)	23(-)	24	25	26	26
38	18(33)	21(36)	23(38)	25(-)	26	27	27	28
40	19(35)	23(38)	25(40)	26(-)	27	28	29	30
42	21(36)	24(40)	26(41)	28(-)	29	30	31	31
44	22(38)	25(41)	28(43)	29(44)	31	31	32	33
46	23(39)	27(43)	29(45)	31(46)	32	33	34	35
48	24(41)	28(45)	31(47)	32(48)	34	35	36	36
50	25(42)	29(46)	32(48)	34(50)	35	36	37	38

- Too few replicates to obtain significance

TABLE 4 (Continued)

$\alpha=0,05$								
n \ k	3	4	5	6		8	9	10
2	-(-)	-(-)	-(-)	-(-)	-(-)	-(-)	-	-
3	-(-)	-(-)	-(-)	-(-)	1(-)	1(-)	1	1
4	-(-)	1(-)	1(-)	1(-)	1(-)	1(-)	1	1
5	1(-)	1(-)	1(-)	2(-)	2(-)	2(-)	2	2
6	1(-)	2(-)	2(-)	2(-)	2(-)	3(-)	3	3
7	2(-)	2(-)	3(-)	3(-)	3(-)	3(-)	3	4
8	2(-)	3(-)	3(-)	4(-)	4(-)	4(-)	4	4
9	3(-)	4(-)	4(-)	4(-)	5(-)	5(-)	5	5
10	3(-)	4(-)	5(-)	5(-)	5(-)	6(-)	6	6
11	4(11)	5(-)	5(-)	6(-)	6(-)	6(-)	6	7
12	5(12)	5(-)	6(-)	6(-)	7(-)	7(-)	7	7
13	5(13)	6(-)	7(-)	7(-)	8(-)	8(-)	8	8
14	6(14)	7(-)	7(-)	8(-)	8(-)	9(-)	9	9
15	6(14)	7(-)	8(-)	9(-)	9(-)	9(-)	10	10
16	7(15)	8(-)	9(-)	9(-)	10(-)	10(-)	10	11
17	7(16)	9(17)	9(-)	10(-)	11(-)	11(-)	11	11
18	8(17)	9(18)	10(-)	11(-)	11(-)	12(-)	12	12
19	8(18)	10(19)	11(-)	12(-)	12(-)	12(-)	13	13
20	9(18)	11(20)	12(-)	12(-)	13(-)	13(-)	14	14
22	10(20)	12(22)	13(-)	14(-)	14(-)	15(-)	15	16
24	11(21)	13(23)	14(24)	15(-)	16(-)	16(-)	17	17
26	13(23)	15(25)	16(26)	17(-)	17(-)	18(-)	18	19
28	14(24)	16(27)	17(28)	18(-)	19(-)	20(-)	20	20
30	15(26)	17(28)	19(30)	20(-)	21(-)	21(-)	22	22
32	16(27)	19(30)	20(31)	21(32)	22(-)	23(-)	23	24
34	17(29)	20(32)	22(33)	23(34)	24(-)	24(-)	25	25
36	18(30)	21(33)	23(35)	24(36)	25(-)	26(-)	27	27
38	20(32)	23(35)	25(37)	26(38)	27(-)	28(-)	28	29
40	21(33)	24(37)	26(38)	27(40)	28(40)	29(-)	30	31
42	22(35)	25(38)	27(40)	29(41)	30(42)	31(-)	32	32
44	23(36)	27(40)	29(42)	30(43)	32(44)	33(-)	33	34
46	25(37)	28(41)	30(44)	32(45)	33(46)	34(-)	35	36
48	26(39)	29(43)	32(45)	34(47)	35(48)	36(48)	37	37
50	27(40)	31(45)	33(47)	35(49)	36(50)	37(50)	38	39